

Lecture 4

Probability and Random Variables

Part 2

RANDOM VARIABLES

Random Variables: From Outcomes to Numbers

A **random variable** assigns a numerical value to each outcome. It allows us to move from qualitative scenarios to quantitative analysis. For the product launch, define:

X =revenue generated by the project

Scenario	Probability p_i	Revenue $X = x_i$
Failure	0.30	0
Moderate success	0.50	50,000
High success	0.20	150,000

Here, X is a **discrete random variable**, because it can take a finite number of possible values.

Probability Distribution of a Discrete Random Variable

A **discrete random variable** assigns a numerical value to each outcome in the sample space.

If the sample space is: $S = \{s_1, s_2, \dots, s_n\}$

each outcome s_i has an associated probability $P(s_i) = p_i$

and the random variable X assigns a numerical value to that outcome: $X(s_i) = x_i$

Often, we associate probabilities with the values of the random variable rather than with the original outcomes."for instance; $P(X = x_i) = p_i$

Although this is not exactly equivalent as we will see in the next exercise.

The probabilities must satisfy:

$$0 \leq p_i \leq 1$$

and:

$$\sum_i p_i = 1$$

Exercise: From Outcomes to Random Variables

Suppose we toss a fair coin three times.

The sample space is: $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

Each outcome has probability:

$$\frac{1}{8}$$

We define two random variables:

- $X=1$ point for each Head and -1 point for each Tail
- $Y=10$ points if all three tosses are equal, and 0 otherwise

Questions

- ☐ Complete the table with the value of X and Y for each outcome.
- ☐ Build the probability distribution of X .
- ☐ Build the probability distribution of Y .

Solution

1. Outcomes, Probabilities and Random Variable Values

Outcome	Prob.	X	Y
HHH	$1/8$	3	10
HHT	$1/8$	1	0
HTH	$1/8$	1	0
HTT	$1/8$	-1	0
THH	$1/8$	1	0
THT	$1/8$	-1	0
TTH	$1/8$	-1	0
TTT	$1/8$	-3	10

2. Probability Distribution of X

Value of X	Prob.
-3	$1/8$
-1	$3/8$
1	$3/8$
3	$1/8$

3. Probability Distribution of Y

Value of Y	Prob.
0	$6/8 = 3/4$
10	$2/8 = 1/4$

To retain: The same sample space can generate different random variables. The outcomes are the same, but X and Y assign different numerical values to those outcomes

Expected Value

The **expected value** of a random variable is its probability-weighted average. For a discrete random variable X , with possible values x_i and probabilities p_i :

$$E[X] = \sum_i p_i x_i$$

Using the product launch example:

$$E[X] = 0.30 \times 0 + 0.50 \times 50,000 + 0.20 \times 150,000$$

$$E[X] = 55,000$$

Outcome	Prob p_i	Revenue x_i
Failure	0.30	0
Moderate success	0.50	50,000
High success	0.20	150,000

The expected value should be interpreted as a probability-weighted average, not as a forecast of what will happen in one specific case. It summarizes the central tendency of the random variable under the assumed probability distribution.

The **expected value** is also called the **mean** of the random variable. It is usually denoted $E[X]$, and sometimes simply μ , read “mu”.

Variance and Standard Deviation

The expected value describes the central tendency of a random variable, similarly to the role of the sample mean in a dataset.

We can also measure the dispersion of a probability distribution. As with datasets, the idea is to measure how far values are from the center; here, the center is the expected value $E[X]$, not the sample mean.

The **variance** summarizes this dispersion.

$$\text{Var}(X) = \sum_i p_i (x_i - E[X])^2$$

The **standard deviation** is the square root of the variance:

$$\sigma_X = \sqrt{\text{Var}(X)}$$

These definitions are very similar to the ones used for datasets. The difference is that, for a random variable, we use **weighted averages**, where the weights are the probabilities attached to the possible outcomes.

Exercise — Expected Value and Variance

We toss a fair coin three times.

Define the random variable **X** as follows:

X=1 point for each Head and -1 point for each Tail

From the previous exercise, the probability distribution of X is:

Value of X	Probability
-3	1/8
-1	3/8
1	3/8
3	1/8

Questions

1. Compute the expected value of X:
2. Compute the variance of X
3. Compute the standard deviation of X

Solution

1. Expected value

$$E[X] = \frac{1}{8}(-3) + \frac{3}{8}(-1) + \frac{3}{8}(1) + \frac{1}{8}(3)$$

$$E[X] = -\frac{3}{8} - \frac{3}{8} + \frac{3}{8} + \frac{3}{8} = 0$$

$$E[X] = 0$$

3. Standard deviation

$$\sigma_X = \sqrt{3}$$

$$\sigma_X \approx 1.73$$

2. Variance

Since $E[X]=0$,

$$\text{Var}(X) = \frac{1}{8}(-3 - 0)^2 + \frac{3}{8}(-1 - 0)^2 + \frac{3}{8}(1 - 0)^2 + \frac{1}{8}(3 - 0)^2$$

$$\text{Var}(X) = \frac{1}{8}(9) + \frac{3}{8}(1) + \frac{3}{8}(1) + \frac{1}{8}(9)$$

$$\text{Var}(X) = \frac{9}{8} + \frac{3}{8} + \frac{3}{8} + \frac{9}{8} = \frac{24}{8} = 3$$

$$\text{Var}(X) = 3$$